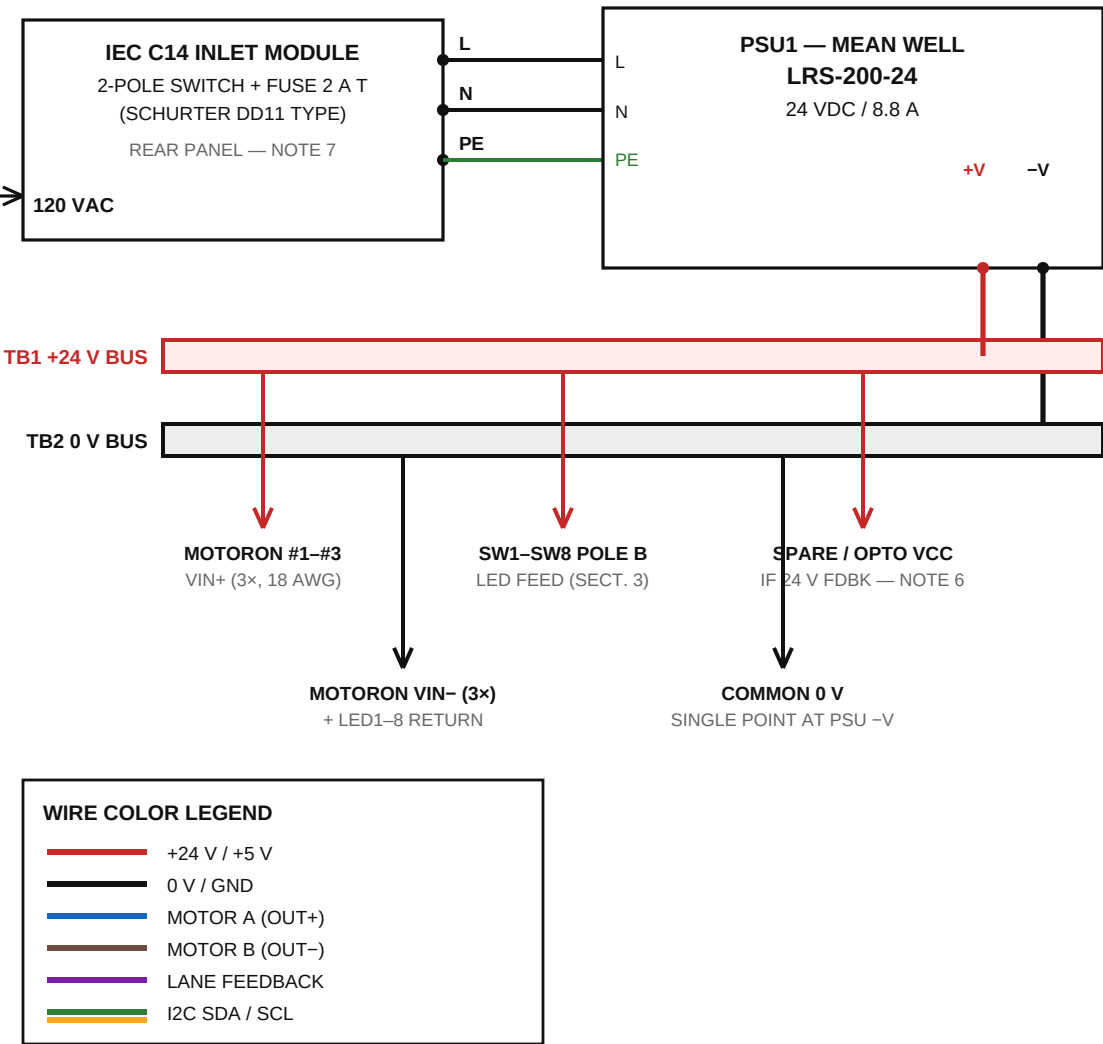
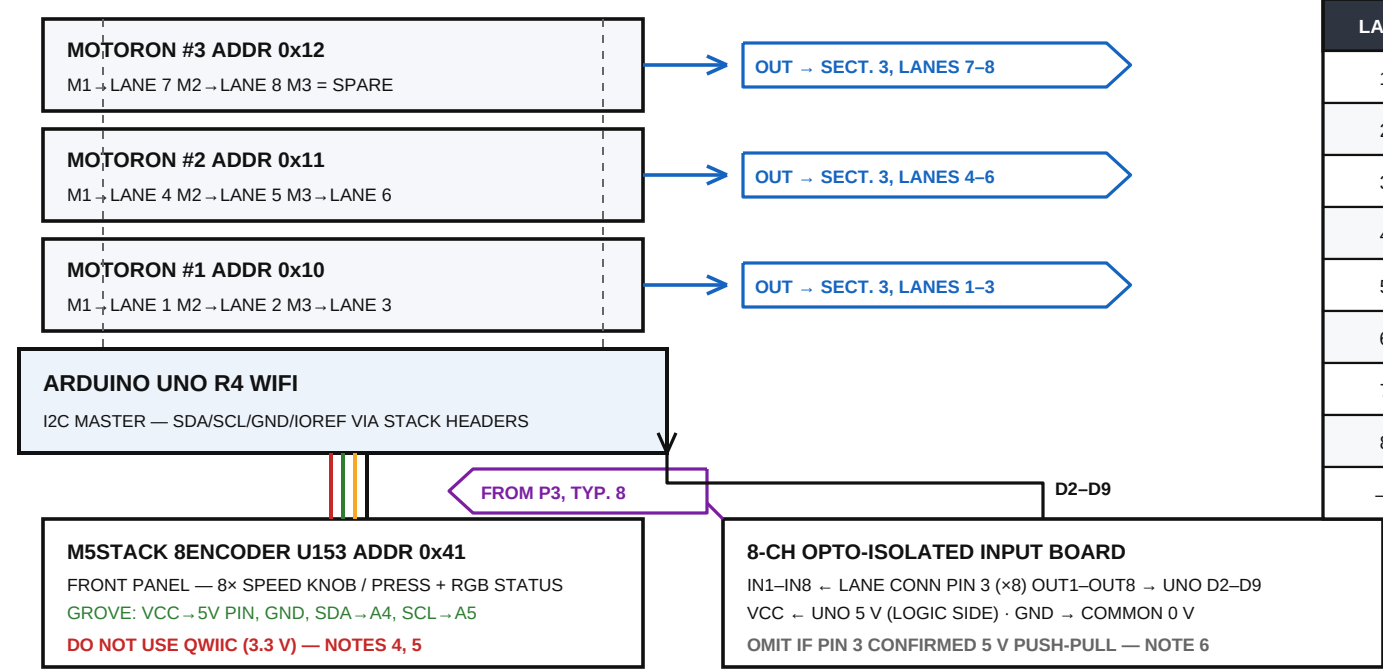


1. POWER INPUT AND DISTRIBUTION



2. CONTROL STACK AND I/O



NOTES

- WIRE: 18 AWG FOR MAINS, 24 V POWER AND MOTOR CIRCUITS; 24 AWG FOR LOGIC AND FEEDBACK. CRIMP FERRULES AT ALL SCREW TERMINALS.
- SW1–SW8: DPST ROCKER WITH GENUINE DC RATING ≥ 6 A @ 24 VDC (BOM ITEM 6). AC-ONLY RATINGS ARE NOT ACCEPTABLE. SWITCH AT STANDSTILL ONLY.
- TVS: BIDIRECTIONAL SMBJ33CA (OR RC SNUBBER 100 Ω + 0.1 μF / 100 V) ACROSS PINS 1–2 AT EACH LANE CONNECTOR, MOTOR SIDE OF SW-n.
- 8ENCODER POWERED FROM UNO 5 V PIN. NEVER FROM QWIC CONNECTOR (3.3 V RAIL).
- VERIFY 8ENCODER 5 V I2C TOLERANCE (OPEN ITEM 8); FIT BIDIRECTIONAL LEVEL SHIFTER ON SDA/SCL IF REQUIRED.
- OPTO BOARD REQUIRED ONLY IF LANE FEEDBACK IS 24 V LEVEL (OPEN ITEM 2). IF PIN 3 IS 5 V PUSH-PULL, LAND PIN 3 DIRECTLY ON D2–D9.
- MAINS: 2 A SLOW-BLOW FUSE IN INLET MODULE. BOND PE TO CHASSIS STUD AND PSU PE TERMINAL. KEEP 120 VAC WIRING CONFINED TO INLET–PSU AREA.
- J-n MATING CONNECTOR P/N PER CONVEYOR SUPPLIER (OPEN ITEM 6). PIN-OUT SHOWN ASSUMED FROM CB002 3-PIN HEADER — CONFIRM BEFORE BUILD.
- MOTORON I2C ADDRESSES SET TO 0x10 / 0x11 / 0x12 BEFORE STACKING. UNO POWERED VIA USB ON THE BENCH; FOR STANDALONE USE FEED VIN FROM A SEPARATE 12 V BUCK — DO NOT FEED VIN FROM THE 24 V BUS.

3. TYPICAL LANE CIRCUIT — LANE n (TYP. 8 PLACES, TABLE 1)

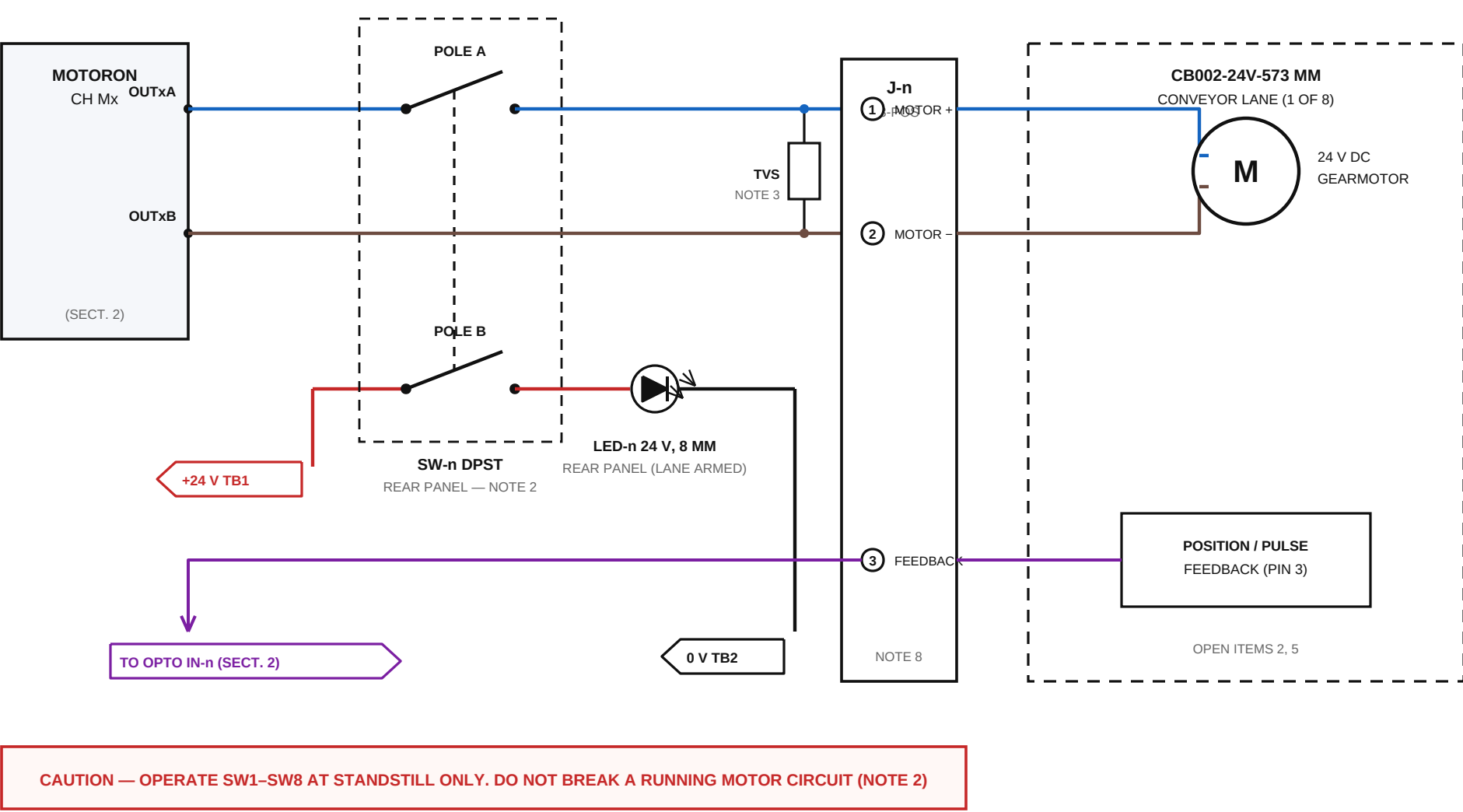


TABLE 1 — CHANNEL ASSIGNMENT

LANE	MOTORON / CH	SWITCH	LED	OPTO IN	UNO PIN
1	#1 · M1	SW1	LED1	IN1	D2
2	#1 · M2	SW2	LED2	IN2	D3
3	#1 · M3	SW3	LED3	IN3	D4
4	#2 · M1	SW4	LED4	IN4	D5
5	#2 · M2	SW5	LED5	IN5	D6
6	#2 · M3	SW6	LED6	IN6	D7
7	#3 · M1	SW7	LED7	IN7	D8
8	#3 · M2	SW8	LED8	IN8	D9
—	#3 · M3	SPARE CHANNEL — NOT WIRED			

REV	DESCRIPTION	DATE
C	8ENCODER CONSOLIDATION (SEE BOM REV C)	08-2026
D	PER-LANE 24 V ISOLATION SWITCH + LED, REAR PANEL	08-25-26
BLACKOCEAN TECHNOLOGY APDU PROGRAM — TEST EQUIPMENT		
APDU LANE DRIVER TEST RIG SHEET 1: WIRING DIAGRAM — TYPICAL LANE CIRCUIT		
SIZE A3	DWG NO. BOT-TR-001	REV D
SHEET 1 OF 2		
DRAWN: U. KAYACIK DATE: 2026-08-25 SCALE: NTS		

CABLE DISTRIBUTION AND HARNESS ROUTING — LANES 1–8

LINES = CABLES. CONDUCTOR-LEVEL DETAIL: SHEET 1, SECT. 3

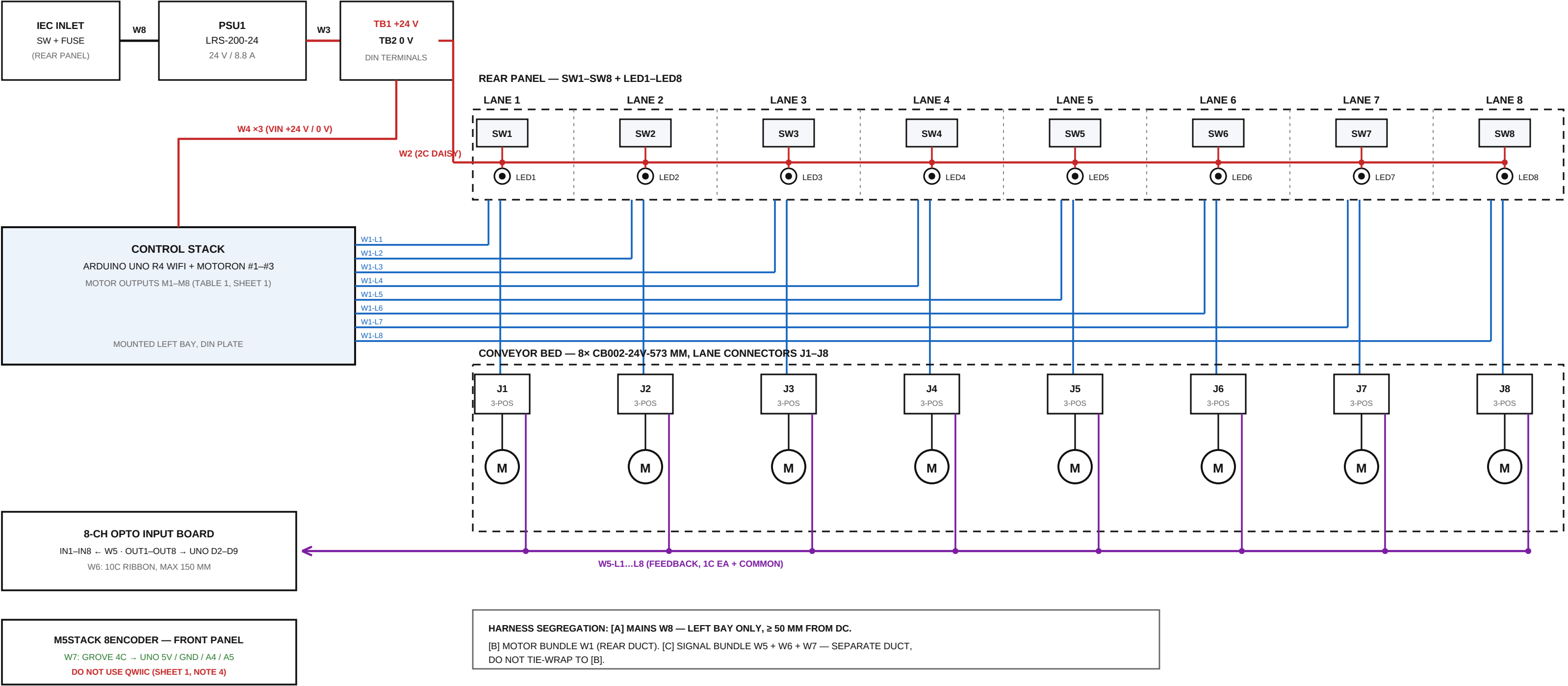


TABLE 2 — CABLE SCHEDULE

CABLE	FROM	TO	COND.	AWG	COLOR	QTY
W8	IEC INLET MODULE	PSU1 L / N / PE	3	18	BRN/BLU/GN-YE	1
W3	PSU1 +V / –V	TB1 / TB2	2	18	RED/BLK	1
W4	TB1 / TB2	MOTORON VIN+ / VIN–	2	18	RED/BLK	3
W1-Ln	MOTORON OUTx (A/B)	SW-n POLE A → J-n P1 / P2	2	18	BLU/BRN	8
W2	TB1 / TB2	SW1–8 POLE B + LED1–8 (DAISY)	2	18	RED/BLK	1
W5-Ln	J-n PIN 3	OPTO IN-n	1+COM	24	VIO	8
W6	OPTO OUT1–8 + 5V/GND	UNO D2–D9	10	24	RIBBON	1
W7	UNO 5V/GND/A4/A5	8ENCODER (GROVE)	4	24	GROVE STD	1

NOTES — SHEET 2

- EACH LINE ON THIS SHEET IS A CABLE, NOT A CONDUCTOR. PIN-LEVEL WIRING: SHEET 1, SECT. 3 AND TABLE 1.
- TWIST W1 MOTOR PAIRS (≥ 10 TURNS/M). ROUTE W5 SIGNAL BUNDLE IN A SEPARATE DUCT FROM W1 — DO NOT MIX.
- LABEL BOTH ENDS OF EVERY CABLE: ID + LANE NO (E.G. W1-L4).
- LEAVE 100 MM SERVICE LOOP AT THE REAR PANEL AND AT J1–J8; STRAIN-RELIEVE AT EACH CONNECTOR.
- W2 DAISY: +24 V AND 0 V LOOP THROUGH ALL 8 SW/LED POSITIONS; TERMINATE FINAL DROP AT SW8.
- ALL CRIMPS FERRULED; TORQUE DIN TERMINALS TO SPEC.
- PE BOND: INLET → CHASSIS STUD → PSU PE (SHEET 1, NOTE 7).

REV	DESCRIPTION	DATE	
C	8ENCODER CONSOLIDATION (SEE BOM REV C)	08-2026	
D	PER-LANE 24 V ISOLATION SWITCH + LED, REAR PANEL	08-25-26	
BLACKOCEAN TECHNOLOGY			
APDU PROGRAM — TEST EQUIPMENT			
APDU LANE DRIVER TEST RIG			
SHEET 2: CABLE DISTRIBUTION — ALL 8 LANES			
SIZE	DWG NO.	REV	SHEET
A3	BOT-TR-001	D	2 OF 2
DRAWN: U. KAYACIK		DATE: 2026-08-25	SCALE: NTS